

Quiz 10

IM/2022/060

IM/2022/017

IM/2022/037

a. Markov Process & CRM

- Modelling Approach (Discrete-Time Markov Chain)
 - Define States: Categorize behavior into mutually exclusive states: In-store, Online, Occasional, and Inactive.
 - Transition Matrix: Use historical purchasing data to build a matrix showing the probability of a customer shifting from one state to another over a specific period.
- Support for Long Term CRM:
 - Steady-State Forecasts: Multiply the matrix to find long-term equilibrium probabilities, predicting the ultimate makeup of the customer base.
 - Scenario Testing: Adjust transition probabilities to simulate CRM initiatives and mathematically forecast their impact on retention and Customer Lifetime Value before investing capital.

b. Queuing Theory (Congestion Analysis)

- System Evaluation:
 - Model the system using arrival rates (λ) and service rates (μ).
 - Calculate key performance metrics: Average queue wait time (W_q), average queue length (L_q), and cashier/server utilization rates (ρ).
- Trade-off Assessment:

- Increasing Staff Capacity: Immediately reduces wait times (W_q), but permanently increases variable costs like wages and creates inefficient idle time (low ρ) during non-peak hours.
- Self-Checkout Tech: Requires high initial Capital Expenditure but lowers long-term variable labor costs. It increases processing channels but introduces risks of customer learning curves and inventory shrinkage like errors and theft.

c. Inventory Allocation via Dynamic Programming

The sequential inventory allocation problem over multiple time periods under uncertainty is a classic scenario for Stochastic Dynamic Programming

- Stages: The multiple time periods, such as days or weeks, over which inventory decisions are made.
 - States: The current inventory levels at the central warehouse and the physical stores at the beginning of each period.
 - Decisions: How many units to allocate to physical stores versus holding at the central warehouse for online fulfillment.
 - Stochastic Element: The uncertain demand for both channels is defined by probability distributions.
- Application: Unlike static optimization, Dynamic Programming works backwards from the end of the planning horizon using Bellman's Equation. It calculates a "policy" rather than a single number. This policy tells LankaMart exactly how much inventory to move, minimizing the total expected combination of holding costs, transfer costs, and stockout penalties over the entire timeline.

d. Game Theory (Competitive Interaction)

- Game Formulation (Two Person Zero Sum):
 - Players: LankaMart (Player 1) and QuickCart (Player 2).
 - Strategies: LankaMart (SL1: Match Price, SL2: Enhance In-Store, SL3: Improve Online) vs. QuickCart (SQ1: Aggressive Price, SQ2: Faster Delivery).
 - Payoff Matrix: Represents the net shift in market share percentage, assuming LankaMart's gain equals QuickCart's exact loss.
- Expected Outcomes & Solution Pathways:
 - Pure Strategy: If LankaMart's Maximin value equals QuickCart's Minimax value, a Saddle Point exists. Both firms consistently use one stable strategy.
 - Mixed Strategy: If no saddle point exists, firms must alternate strategies based on calculated probabilities. Because LankaMart has 3 strategies and QuickCart has 2, LankaMart would use the Graphical Method or Linear Programming (Simplex Method) to solve for the optimal strategic mix.

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Q
2.

AlpaTech/ BetaMobile	X (Premium)	Y (Balanced)	Z (Discount)	
A: Premium	8	2	6	2
B: Balanced	3	7	5	3 ← maxmin Value
C: Discount	4	6	1	1
	8	7	6	6 ↑ minmax Value

Since minmax value $\neq$ maxmin value

$$6 \neq 3$$

$\therefore$ there is no saddle point.

Checking if there are any reductions can be done by applying dominance rules.

Row A is not always better than B or C

Row B is not always better than A or C

Column X is not always worse than Y or Z

$\therefore$ Matrix remains 3×3

The Goal - Maximize the value of the game (V) for AlpaTech

x_1, x_2, x_3 - Probabilities of Alpha Tech choosing A, B, C

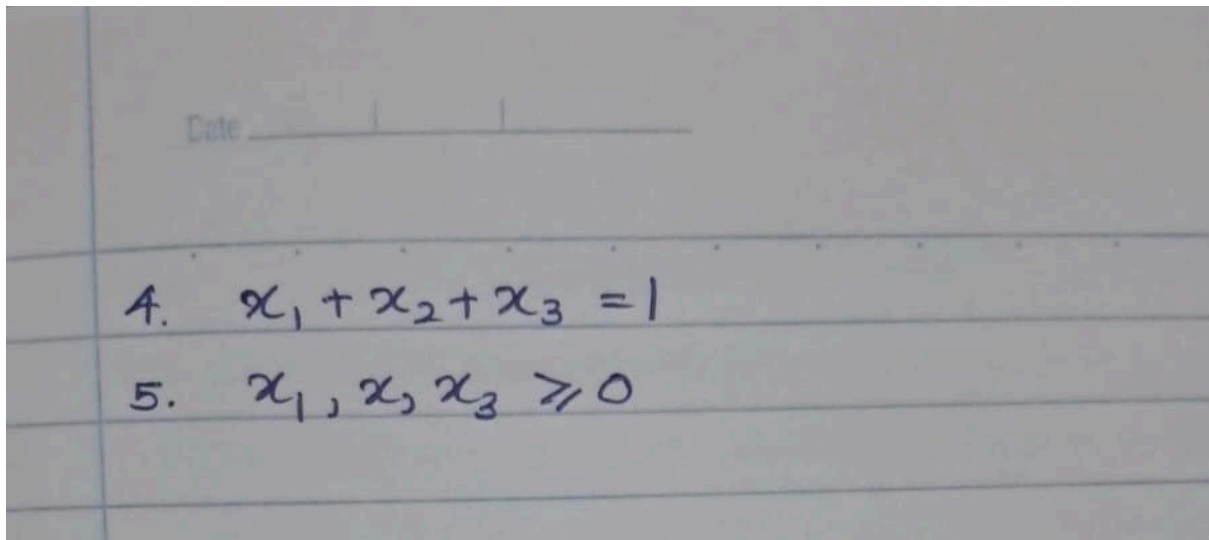
V - The value of the game

The Constraints.

1. For X: $8x_1 + 3x_2 + 4x_3 \geq V$

2. For Y: $2x_1 + 7x_2 + 6x_3 \geq V$

3. For Z: $6x_1 + 5x_2 + 1x_3 \geq V$



3.

First, we must determine if a stable pure strategy exists by finding the Maximin and Minimax values.

- Row Minimums (AlphaTech's worst-case payoffs):
 - Row A (Premium): 2
 - Row B (Balanced): 3 (Maximin)
 - Row C (Discount): 1
- Column Maximums (BetaMobile's worst-case losses):
 - Column X (Premium): 8
 - Column Y (Balanced): 7
 - Column Z (Discount): 6 (Minimax)

Conclusion: Since the Maximin (3) does not equal the Minimax (6), there is no saddle point.

The companies must use mixed strategies.

Step 2: Matrix Reduction using the Rule of Dominance Before solving, we can simplify the 3x3 matrix by eliminating inferior strategies.

- Eliminating Row C (AlphaTech): Strategy C (Discount) is strictly dominated by a mixed combination of Strategy A (20%) and Strategy B (80%).
 - Expected payoffs of the mix: $X = 4$, $Y = 6$, $Z = 5.2$.
 - Since these expected payoffs (4, 6, 5.2) are greater than or equal to Row C's direct payoffs (4, 6, 1), AlphaTech will never rationally choose Strategy C. Row C is eliminated.
- Eliminating Column Z (BetaMobile): Looking at the remaining 2x3 matrix, Strategy Z (Discount) is dominated by a 50/50 mix of Strategy X and Strategy Y.
 - Expected losses of the mix: $A = 5$, $B = 5$.
 - Since these expected losses (5, 5) are less than or equal to Column Z's direct losses (6, 5), BetaMobile will never choose Strategy Z. Column Z is eliminated.

Reduced 2x2 Matrix:

- Row A: 8, 2

Solving the 2x2 Matrix

- AlphaTech's Probabilities (p for A, $1-p$ for B): Equating the expected payoffs against BetaMobile's strategies: $8p + 3(1 - p) = 2p + 7(1 - p)$
 $5p + 3 = -5p + 7$
 $10p = 4$
 $p = 0.4$
 (Therefore, $1-p = 0.6$)
- BetaMobile's Probabilities (q for X, $1-q$ for Y): Equating the expected losses against AlphaTech's strategies: $8q + 2(1 - q) = 3q + 7(1 - q)$
 $6q + 2 = -4q + 7$
 $10q = 5$
 $q = 0.5$
 (Therefore, $1-q = 0.5$)
- Value of the Game (V): Substituting AlphaTech's probability ' p ' back into the expected payoff equation: $V = 5(0.4) + 3$
 $V = 5$

- Value of the Game: 5 (*This game strictly favors AlphaTech, guaranteeing them an expected payoff/gain of 5 units per cycle if played optimally*).
- AlphaTech's Optimal Strategy: (0.4, 0.6, 0) (*AlphaTech should utilize the Premium strategy 40% of the time, the Balanced strategy 60% of the time, and completely abandon the Discount strategy*).
- BetaMobile's Optimal Strategy: (0.5, 0.5, 0) (*BetaMobile should respond with a 50/50 mix of their Premium and Balanced strategies, completely abandoning the Discount strategy to minimize their losses*).